

# DS 669 - Reinforcement Learning

## Programming Assignment #0

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FrozenLake-v1 Environment Initialization, Functions, and Testing

### II. Environment Initialization

#### II(a). Default 4x4 Environment

After running `code_base.py` without changing the default settings, I obtained a 4x4 FrozenLake-v1 environment. The environment opened in a separate window and the agent began at the starting position.



Figure 1. FrozenLake-v1 initialized with the default 4x4 map.

#### II(b). 8x8 Environment

I changed the `map_size` default value from `'4x4'` to `'8x8'` in `get_args.py`. After running `code_base.py`, the FrozenLake-v1 environment was successfully initialized with an 8x8 map.

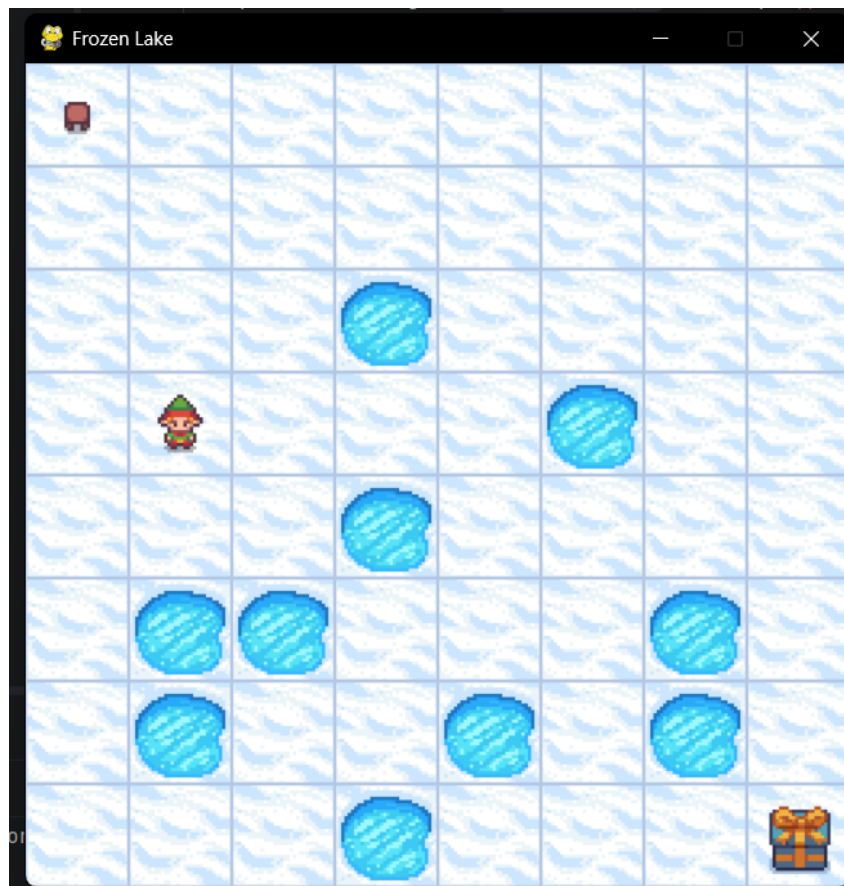


Figure 2. FrozenLake-v1 initialized with the 8x8 map.

### III. Environment Functions

#### III(a). Initial State

The initial state is **state 0**. Its location on the 8x8 map is **[0, 0]**, which is the top-left corner of the map.

#### III(b). Result of Action 1

At the initial state, action 1 moves the agent down. The next state is **state 8**, which is located at **[1, 0]**. The reward is 0.0 and the episode does not terminate.

```
mode: test_action
0
action:0, next state:0, reward:0.0, done:False, prob: {'prob': 1.0}
action:1, next state:8, reward:0.0, done:False, prob: {'prob': 1.0}
action:2, next state:1, reward:0.0, done:False, prob: {'prob': 1.0}
action:3, next state:0, reward:0.0, done:False, prob: {'prob': 1.0}
```

Figure 3. Console output from test\_action showing the initial state and results of actions 0-3.

### IV. Environment Test

#### IV(a). Default Action Sequence

Yes. The agent reached a terminal state and the episode ended. After executing the provided default action sequence, the final output showed done=True at **state 52**. The total reward was **0.0** and the sequence used **16 steps**.

#### IV(b). Shortest Path to the Goal

The minimum number of steps required to reach the goal from the starting position is **14 steps**. I used seven right actions followed by seven down actions. The action list was **[2, 2, 2, 2, 2, 2, 2, 1, 1, 1, 1, 1, 1, 1]**. The agent reached **state 63**, received a reward of **1.0**, and the episode terminated successfully with done=True.

```
mode: test_moves
next state:1, reward:0.0, done:False
next state:2, reward:0.0, done:False
next state:3, reward:0.0, done:False
next state:4, reward:0.0, done:False
next state:5, reward:0.0, done:False
next state:6, reward:0.0, done:False
next state:7, reward:0.0, done:False
next state:15, reward:0.0, done:False
next state:23, reward:0.0, done:False
next state:31, reward:0.0, done:False
next state:39, reward:0.0, done:False
next state:47, reward:0.0, done:False
next state:55, reward:0.0, done:False
next state:63, reward:1.0, done:True
total reward:1.0, num_steps: 14

Process finished with exit code 0
```

Figure 4. Console output showing the 14-step shortest path reaching state 63 with reward 1.0.